

HIRA AHMED

AI Engineer | Backend & Automation | QA Lead

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PROFESSIONAL SUMMARY

AI Engineer specializing in production AI systems, automation, backend engineering, and software quality. Experienced in building LLM pipelines, AI agents, n8n automations, API integrations, and AWS-backed systems, with a strong focus on reliability, failure handling, and verification. Built and operated production systems processing 28,000+ records, improved pipeline throughput by 5.6×, and reduced monthly AWS infrastructure costs by 34%. Also leads QA across multi-tenant B2B software, testing end-to-end business workflows, APIs, authorization, data integrity, and security-sensitive edge cases. Lead author and developer of Sentinel-Mesh, a neuro-symbolic research framework combining LLM-generated Terraform remediation with Z3 SMT formal verification.

PROFESSIONAL EXPERIENCE

TechPotion.ai, Remote · Malaysia-based

Jan 2025 – Present

Quality Assurance Lead

Sep 2026 – Present

QA Lead across three simultaneous client contracts, owning end-to-end defect discovery, API-level evidence, authorization, multi-tenant isolation, and business-logic validation.

PharmaConnect · QA Contract

- Reduced critical defect escape rate by leading QA across 20 end-to-end journeys and 6 user roles, uncovering 11 critical P1 findings out of 80+ total documented defects across 15 modules.
- Eliminated authorization and isolation regressions by testing authentication, RBAC, tenant isolation, pricing, financial reconciliation, audit logging, and data integrity with full API reproduction evidence.
- Accelerated remediation by tracing recurring defects to shared root causes, cutting duplicate fix cycles and improving regression coverage across the platform.

ToolPotion · QA Contract

- Improved production listing quality for a 28K+ AI tools directory by documenting 100+ defects across relevance, data integrity, rendering, localization, and leaderboard/category behavior.
- Protected platform credibility by identifying 190+ dead or fraudulent listings in a targeted production pass and surfacing security risks among live listings.
- Validated multilingual and publishing workflows end-to-end across two full QA cycles.

AI Cloud Academy · QA Contract

- Eliminated a gamification exploit by identifying and documenting an XP/progress vulnerability, then validating the fix through controlled regression testing across 35 documented defects.
- Ensured learner-journey reliability by testing progress-tracking, data integrity, and end-to-end business logic across all relevant user flows.

Backend & Automation Engineer

Jan 2025 – Sep 2026

- Increased pipeline throughput by 5.6× across 28,000+ records by re-architecting ETL into a fault-tolerant parallel system with 50 concurrent workers, dynamic rate limiting, and circuit breakers.
- Reduced monthly AWS infrastructure costs by 34% through query optimisation and execution-flow refactoring across Lambda, RDS, EC2, S3, API Gateway, and EventBridge.
- Achieved ~95% data-extraction success rate by building Python scrapers (BeautifulSoup, Playwright, Selenium) across 10 platforms with 20+ extraction methods and a URL-healing service.
- Enabled 100% schema-valid LLM output for categorisation, FAQ generation, and 9-language translation by enforcing JSON-mode on a 7-stage status-gated enrichment pipeline via OpenRouter.
- Shipped 12+ production automation workflows using n8n and Make.com alongside FastAPI, PostgreSQL, Docker, Next.js, and LLM APIs.

Sadiq.ai, Karachi, Pakistan

Jun 2024 – Aug 2024

QA Intern

- Improved dev-team bug-query efficiency by ~40% by building a RAG-based internal chatbot (Gemini API, Python, Streamlit) to surface defects from tracking data in real time.
- Reduced release-blocking defects by documenting 150+ edge-case bugs across web and mobile through manual and automated Flutter testing (unit, widget, golden tests).

RESEARCH

Sentinel-Mesh: Formally Verified Remediation of Cloud Misconfigurations · Lead Author & Developer · 2026

Neuro-symbolic framework combining LLM-generated Terraform remediation with Z3 SMT formal verification. Evaluated on CloudFix-Bench (105 AWS Terraform cases): achieved 83.81% remediation rate (88/105), 0.0% security regression, and 29 formal proof certificates. Preprint on Research Square (DOI: 10.21203/rs.3.rs-10674271/v1); under review at **IEEE Transactions on Cloud Computing** (Manuscript ID TCC-2026-08-0693). Artifact archived on Zenodo (DOI: 10.5281/zenodo.20975067).

SELECTED PROJECTS

Autonomous Cloud Compliance & AI Auditing Engine

[LangGraph](#) · [CRAG](#) · [LlamaIndex](#) · [Supabase pgvector](#) · [FastAPI](#) · [n8n](#) · [AWS](#)

- Cut questionnaire turnaround by ~90% by engineering an autonomous engine that auto-answers 250+ question enterprise reviews (CAIQ, SOC 2, SIG) from live AWS infrastructure evidence.

- Eliminated LLM hallucinations (100% citation enforcement) via a 6-node LangGraph CRAG state machine with Llama-3.1-8B relevance scoring and an automated Critic Agent verifying real AWS ARNs.
- Achieved 0.95 semantic relevance score at \$0 embedding cost using PyMuPDF + FastEmbed all-MiniLM-L6-v2 stored in Supabase vector(384); processed 2,000+ question workloads with zero timeout failures.

Autonomous AI Email Agent

[n8n](#) · [Gmail API](#) · [Groq](#) · [RAG](#) · [Google Sheets](#)

- Eliminated manual screening overhead by building a stateful email agent with thread-level memory for automated candidate screening and technical Q&A,, with explicit branching for malformed inputs.

Autonomous Competitor Intelligence & SEO Pipeline

[n8n](#) · [ClickUp](#) · [Gemini](#) · [PostgreSQL](#)

- Automated competitor research, SEO gap analysis, and brief generation by building a PostgreSQL-backed pipeline with dead-letter queue patterns ensuring 100% data integrity on concurrent updates.

TECHNICAL SKILLS

AI & LLM Systems	LangGraph, LangChain, LlamaIndex, RAG/CRAG, LLM pipelines, structured outputs, Groq, Gemini Flash, GPT-4o, Claude 3.5, OpenRouter, prompt engineering, pgvector, FastEmbed
Automation & Orchestration	n8n, Make.com, async state machines, concurrency locks, rate limiting, dead-letter queues, exponential backoff, idempotency, replay patterns
Backend & APIs	Python (advanced), FastAPI, Next.js 15, REST APIs, webhooks, OAuth, TypeScript, JavaScript, SQL
Cloud & Infrastructure	AWS (Lambda, RDS, EC2, S3, API Gateway, EventBridge, SQS, CloudWatch, IAM, VPC, CloudFormation), Terraform, Docker Compose, CI/CD, Linux/Bash
Databases	PostgreSQL (JSONB, GIN indexing, pgvector), Supabase, DynamoDB, NoSQL
QA & Reliability	Multi-tenant B2B QA, API testing, RBAC/authorization, business-logic testing, data-integrity testing, exploratory testing, regression testing, browser/network inspection
Security & Formal Verification	Z3 SMT solving, cloud security invariants, Terraform security, IAM policy analysis, security testing

EDUCATION & CREDENTIALS

B.S. Cybersecurity

Sir Syed University of Engineering and Technology (SSUET), Pakistan
Top 10 Finalist (Blue Team), Ignite National Hackathon (200+ teams)

Credentials & Open Source

ISC2 Certified in Cybersecurity (CC)
Verified Peer Reviewer, IEEE Access (5 reviews, Web of Science / ORCID)
[n8n Production Resilience Patterns](#) [github.com/hira299/n8n-production-resilience-patterns](#)
[Enterprise Multi-Tenant QA Matrix](#) [github.com/hira299/enterprise-multitenant-qa-matrix](#)